

5 Many animals have a heart and circulation.

(a) Molecules such as glucose move by mass transport to overcome the limitations of diffusion in meeting the requirements of an animal.

(i) State what is meant by the term **mass transport**.

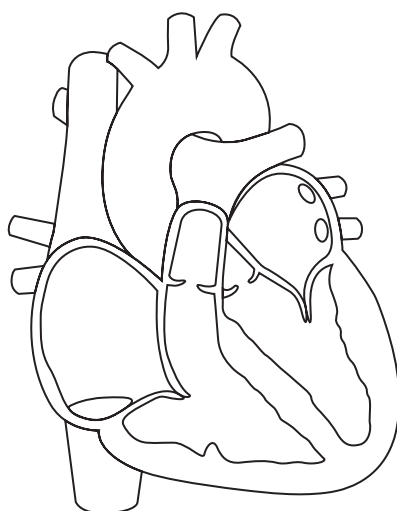
(1)

(ii) All the cells in an animal require glucose.

Explain the limitations of diffusion in providing cells with glucose.

(2)

(b) The diagram shows a mammalian heart.



Draw arrows on the diagram to show the flow of deoxygenated blood (oxygen-depleted blood) into, through and out of the heart.

(2)

(c) The tunica media is the middle layer in the wall of an artery.

The table shows the percentages of some components in the tunica media of arteries of different sizes.

Component	Percentages of some components in tunica media (%)		
	Large artery	Medium-sized artery	Small artery
Smooth muscle cells	33.5%		60.5%
Collagen	37.0%		12.0%
Elastin	24.5%		1.5%

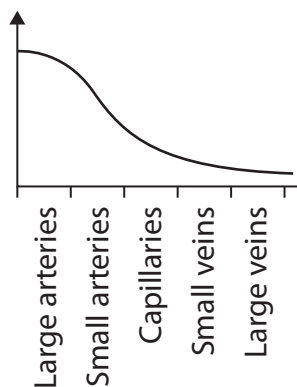
Explain why the percentages of the components in these arteries are different.

(4)

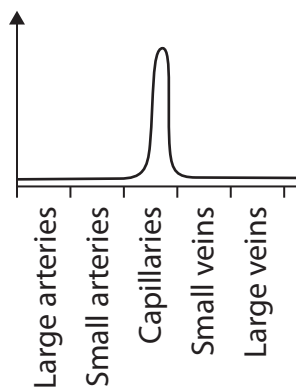


(d) The graphs show differences in some features of blood vessels.

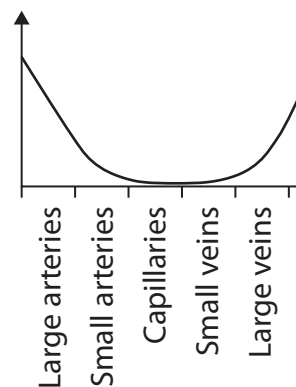
Graph 1



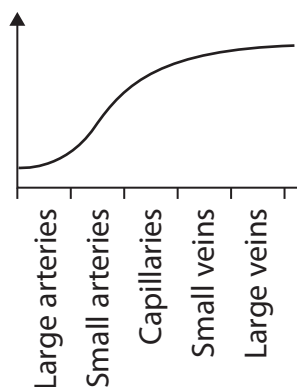
Graph 2



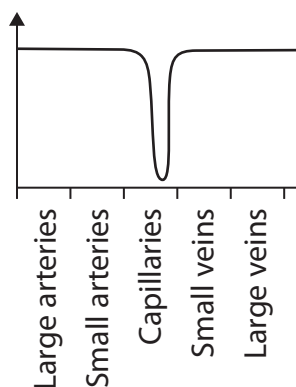
Graph 3



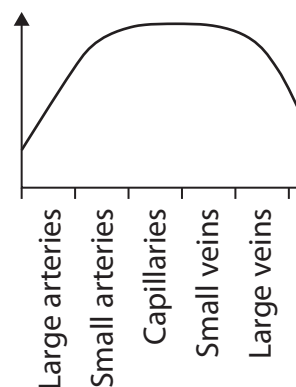
Graph 4



Graph 5



Graph 6



(i) Which graph shows the diameter of the blood vessels?

(1)

- ☐ **A** graph 1
- ☐ **B** graph 3
- ☐ **C** graph 4
- ☐ **D** graph 6

(ii) Which graph shows the permeability of the blood vessels?

(1)

- ☐ **A** graph 1
- ☐ **B** graph 2
- ☐ **C** graph 3
- ☐ **D** graph 6

(Total for Question 5 = 11 marks)